# COLLEGE PROCUREMENT MANAGEMENT SYSTEM

**A PROJECT REPORT**

*Submitted by*

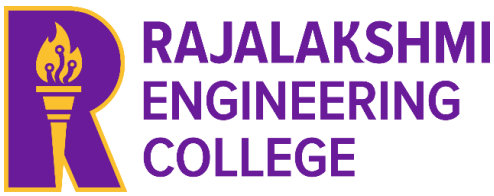
**ARJUNRAJ M (240701049)**

*in partial fulfilment of the End Semester Examination for the course*

**CB23F35 –**
**LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR**

*in*

**BACHELOR OF ENGINEERING**
**COMPUTER SCIENCE AND ENGINEERING**




**RAJALAKSHMI ENGINEERING COLLEGE**
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

**NOVEMBER 2026**

# BONAFIDE CERTIFICATE

**RAJALAKSHMI ENGINEERING COLLEGE**
(An Autonomous Institution, Affiliated to Anna University, Chennai)

Certified that this project report titled "College Procurement Management System" is the bonafide work of **ARJUNRAJ M** (240701049), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on ____ Nov, 2026

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# DECLARATION

I, ARJUNRAJ M (240701049), hereby declare that the project report titled "College Procurement Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard page, Deluge scripts, workflows, approval process and blueprint — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: ____-11-2026

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# ACKNOWLEDGEMENT

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I owe my sincere thanks to my course faculty, Dr. JINU SOPHIA, Assistant Professor, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and its learning resources, and I thank my family and friends for their constant support.

**ARJUNRAJ M**

# ABSTRACT

Colleges buy equipment, software, stationery and maintenance services throughout the year. In many institutions a purchase request still travels as a paper form from the requesting staff member to the Head of the Department (HOD) and then to the finance office. Budget availability is checked by hand, approvals are given by signature, and nobody can easily see which request is pending, approved or rejected. This causes delay, duplicate data entry, overspending risk and no reliable record of who approved what.

The College Procurement Management System is a low-code application built on Zoho Creator to solve this problem. It manages the purchase life cycle in six connected forms: Department, Budget, Vendor, Purchase_Request, Purchase_Request_Items (used as a line-item subform) and Approval_History. Deluge workflows calculate the estimated total of a request from its line items, compare it with the available budget of the department and budget head, move the request through the HOD and Finance approval stages, and write an entry in the approval history at every decision. An approval process sends an email to the requester when the HOD approves or rejects a request, and an approval blueprint defines the stages Submitted, HOD Review, Finance Review, Procurement and Approved. Eight list reports and the Procurement Dashboard (KPI boxes, a pie chart, a column chart, a budget overview and embedded work-lists) give each user a live view of the process. Role-based profiles for Administrator, HOD, Finance and Requester, together with portal profiles, give every user only the data and actions that belong to the job.

The application replaces paper forms and spreadsheets with one online system, reduces repeated data entry, and gives the HOD, the finance officer and the requester a live view of every request from submission to approval.

**Keywords:** Low-Code Development; Zoho Creator; Deluge Scripting; Approval Workflow; Budget Control; Role-Based Access Control; Procurement Management

# PROJECT AT A GLANCE

| Attribute | Details |
|---|---|
| Project Title | College Procurement Management System |
| Application Domain | Education / Administration – purchase request, budget check and multi-level approval |
| Student Name & Register No. | ARJUNRAJ M – 240701049 |
| Zoho Creator Edition | Trial (screens show “Trial expires in 14 days”) |
| Application Name (Link Name) | College |
| Published / Live URL | [paste the live application URL here] |
| Demo Credentials (per role) | [Administrator \| HOD \| Finance \| Requester – add user names only; never write passwords here] |
| Number of Forms / Subforms | 6 / 1 (Items subform in Purchase_Request, based on Purchase_Request_Items) |
| Number of Reports (by type) | List: 8  Chart: 0 (two charts are placed on the dashboard page)<br>Kanban: 0  Calendar: 0  Pivot: 0 |
| Number of Pages/ Dashboards | 1 (Procurement_Dashboard) |
| Number of Workflows | Form: 5  Schedule: 0  Approval: 1  Blueprint: 1 (plus 1 empty draft)  Batch: 0 |
| Deluge Scripts / Functions | 5 scripts (DS-01 to DS-05) / 0 custom functions |
| Integrations | Zoho Mail (email notifications through the sendmail task of the approval process) |
| Roles & Profiles Defined | Roles: CEO (top), Finance, HOD, Requester. User profiles: Administrator, HOD, Finance, Requester, Read, Write, Developer. Portal profiles: HOD, Finance Portal, Requester Portal, Customer |
| Environments Used | Development → Production (strategy described in Chapter 7) |
| Date of Production Deployment | [dd-mm-2026] |

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# LIST OF ABBREVIATIONS

| Abbreviation | Expansion |
|---|---|
| API | Application Programming Interface |
| CRUD | Create, Read, Update, Delete |
| DS | Deluge Script |
| ER | Entity–Relationship |
| FR | Functional Requirement |
| HOD | Head of the Department |
| INR / USD | Indian Rupee / United States Dollar (currency field types) |
| KPI | Key Performance Indicator |
| LCAP | Low-Code Application Platform |
| NFR | Non-Functional Requirement |
| PR | Purchase Request |
| RBAC | Role-Based Access Control |
| REST | Representational State Transfer |
| SDLC | Software Development Life Cycle |
| UAT | User Acceptance Testing |
| UI | User Interface |
| Zia | Zoho’s AI assistant |

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

A college needs a steady supply of goods and services: desktop computers and projectors for laboratories and classrooms, software licences for students, notebooks and office stationery, and repair of existing equipment. Each department asks for what it needs, but the money available is limited. The college therefore keeps a yearly budget for every department, divided into budget heads such as Equipment, Stationery, Software and Maintenance. Before an item is bought, the request must be approved by the Head of the Department (HOD) and then by the finance office, who confirm that money is available under the right budget head.

This project is built for a college (a realistic case study) where many departments raise purchase requests every month. The people involved are the requester, who is a faculty or staff member raising the request; the HOD, who reviews whether the purchase is needed; the finance officer, who checks the budget and gives the final financial approval; and the administrator, who maintains departments, budgets and vendors. At present most of this work is done on paper and in spreadsheets, which makes the process slow and difficult to track.

## 1.2 Problem Statement

Purchase handling in the college is not managed on one common system. Requests, budgets, vendor details and approval notes are kept in separate files and registers. Because of this, approvers spend time searching for information, requesters do not know where their request is stuck, and management cannot see how many requests are pending, approved or rejected. A single online system is needed that connects every step of a purchase request and keeps all data in one place. The main pain points are:

- The same department, item and amount details are typed again in different spreadsheets, which causes errors and duplicate data.
- The total of a request is calculated by hand, and the budget is checked manually, so a request can be approved even when the budget is not enough.
- Approvals are taken by signature or message, and there is no record of who approved or rejected a request, when, and with what comments.
- A request that needs correction has no clear “revision required” path, so it is lost or delayed.
- Requesters and managers have no real-time view of request status, and reports need manual work.

## 1.3 Objectives

- To design a normalised data model and forms for departments, budgets, vendors, purchase requests, request items and approval history.
- To automate total calculation, budget checking, status change and approval-history logging using Deluge scripting and workflows.

- To implement a two-level approval flow (HOD and Finance) with an approval process, an approval blueprint and decision fields that are visible only to the right profile.
- To provide role-based access, list reports and a dashboard so that each user sees only the information needed for the job.
- To send an email notification to the requester when the HOD approves or rejects a request, using Zoho Mail through the sendmail task.
- To make the application usable on web and mobile browsers and to prepare it for deployment to a production environment with governance controls.

## 1.4 Scope of the Project

**In scope:** maintenance of departments, budgets and vendors; creation of purchase requests with multiple line items and supporting documents; automatic calculation of the estimated total; budget check by department and budget head; HOD and Finance decisions with comments; approval history; eight list reports; the Procurement Dashboard; role-based access for four user roles with portal profiles; email notification for HOD decisions; and access through a web browser and a mobile browser.

**Out of scope:** automatic creation of purchase orders for vendors, vendor bidding and quotation comparison, goods-received and invoice matching, integration with a real accounting or payment system, and automatic updating of the Reserved and Spent amounts of a budget. These items are listed as future enhancements in Chapter 9.

## 1.5 Justification for a Low-Code Approach

A procurement system needs many forms, relationships between data, multi-level approvals, notifications and reports. Building this from scratch with traditional programming would need a database design, a back-end, a front-end, hosting and a separate mobile application. Low-code platforms reduce this effort because the common parts are available as ready-made building blocks [4], [5]. Zoho Creator provides drag-and-drop form building, built-in reports and pages, a simple scripting language (Deluge), approval and blueprint workflows, cloud hosting, role-based security and mobile access in one place [1], [2]. This allows the application to be built quickly, changed easily when approval rules change, and maintained by a citizen developer. The cost of servers and maintenance is also avoided because Zoho hosts the application.

## 1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

# CHAPTER 2
# LITERATURE SURVEY AND SYSTEM STUDY

## 2.1 Study of the Existing System

In the existing system, a requester writes a paper purchase form or informs the office by phone. A clerk copies the details into a spreadsheet. The form is carried to the HOD for signature and then to the finance office, where the budget is checked in a separate register. If the requester wants to know the status, he or she must call or visit the office. The flow is shown below.

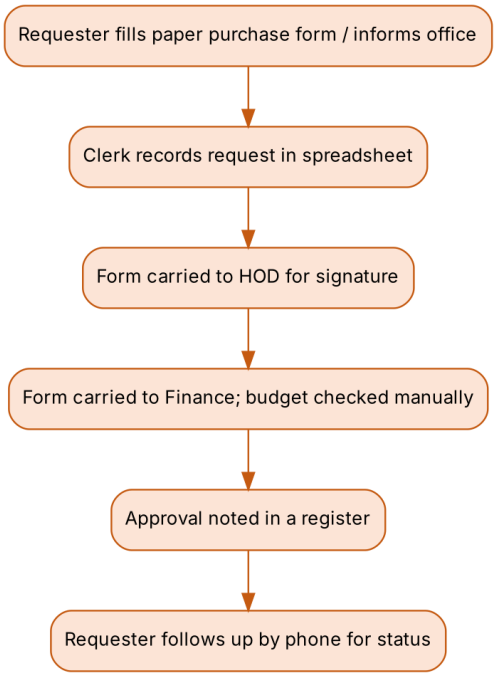



**Figure 2.1:** As-Is Process Flow of the Existing System

## 2.2 Limitations of the Existing System

- Data is entered more than once, so errors and mismatches are common.
- The total of a request and the available budget are calculated manually, so overspending can go unnoticed.
- Approval history is not recorded in a reliable way, and there is no audit trail of who approved a request and when.
- There is no defined path for “revision required”, so corrected requests are easily lost.
- Request status cannot be seen online, and reports need a lot of manual effort.
- Anyone who has the spreadsheet can see all departments ’budgets and requests, so security is weak.

## 2.3 Low-Code versus Traditional Development

The table below compares the two approaches for this application. The effort values are estimates based on the size of this project (six forms, five scripts, five form workflows, one approval process and one dashboard).

**Table 2.1:** Comparison of Low-Code and Traditional Development for the Proposed Application

| Criterion | Traditional (Pro-Code) Development | Low-Code on Zoho Creator |
|---|---|---|
| Development time | About 8–10 weeks (database, backend, front-end, testing) | About 2–3 weeks (forms, scripts and workflows configured) |
| Skills required | Back-end, front-end and database developers | One citizen developer with basic Deluge knowledge |
| Hosting & infrastructure | Own server or cloud set-up, backups and monitoring to be managed | Hosted and backed up by Zoho; nothing to install |
| Mobile application | Separate Android/iOS development needed | Available through the Zoho Creator mobile app and mobile browser |
| Security & access control | Login, roles and permissions coded manually | Built-in users, roles, profiles and portal profiles |
| Maintenance & change | Code change, testing and redeployment for each change | Form or workflow is edited visually and published |
| Customisation limits | Almost no limit | Limited by platform features and API limits; extended with Deluge |
| Cost | High (developer time, servers, licences) | Low (platform subscription; trial and education editions available) |

## 2.4 Literature Survey

The following sources were studied to understand low-code development and to decide the design of the application.

**Table 2.2:** Summary of Literature Survey

| S.No. | Author(s) / Source & Year | Title / Product | Key Findings | Relevance to this Project |
|---|---|---|---|---|
| 1 | Zoho Corporation, online documentation [1] | Zoho Creator Help Documentation | Describes forms, reports, pages, workflows, approvals, blueprints and user management. | Main reference for building every module. |
| 2 | Zoho Corporation, online documentation [2] | Deluge Script Reference Guide | Gives syntax for fetch, insert, validation, loops and sendmail. | Used for DS-01 to DS-05 and the approval actions. |
| 3 | A. C. Bock and U. Frank, 2021 [4] | Low-Code Platform | Defines low-code platforms and their main characteristics and limits. | Used to justify the approach and to discuss limits in Chapter 8. |

| S.No. | Author(s) / Source & Year | Title / Product | Key Findings | Relevance to this Project |
|---|---|---|---|---|
| 4 | R. Sanchis et al., 2020 [5] | Low-Code as Enabler of Digital Transformation in Manufacturing Industry | Shows that low-code speeds up digital transformation for small and medium organisations. | Similar benefit expected for a college administration office. |
| 5 | A. Sahay et al., 2020 [6] | Supporting the understanding and comparison of low-code development platforms | Compares platforms using common features such as data model, workflow and integration. | Helped to check that Zoho Creator covers the required features. |

## 2.5 Emerging Trends in Low-Code Development

Several trends are shaping low-code development. Citizen development allows business users to build their own applications without waiting for a large IT team. AI-assisted building lets a developer describe an application in plain words and receive a first version of forms and workflows, which can then be improved. Hyperautomation joins approvals, scripts, notifications and integrations into one automatic flow. Composable applications are built from small reusable parts that can be changed independently. The College application follows these ideas: each master and transaction is a separate form, the forms are joined by lookup fields and workflows, and the same design can be extended later with AI features such as quotation reading or spend analysis.

## 2.6 Proposed System

The proposed system is the College application on Zoho Creator. It keeps all data in one cloud database and connects departments, budgets, requests, items and approval history through lookup fields, so details are entered only once. Workflows calculate the estimated total from the line items and compare it with the available budget of the department and budget head. The HOD and Finance record their decisions in the request itself; each decision moves the request to the next status and writes a row in the approval history. An approval process and an approval blueprint define the review stages, and an email is sent to the requester when the HOD approves or rejects. Role-based profiles protect the decision fields and reports, and the dashboard gives a live view of the process. In this way every limitation listed in Section 2.2 is addressed.

# CHAPTER 3
# REQUIREMENT ANALYSIS

## 3.1 Stakeholders and User Roles

**Table 3.1:** Stakeholders and User Roles

| Role | Description | Key Responsibilities in the Application |
|---|---|---|
| Administrator | Owner of the application and its configuration | Manage users, roles and profiles; maintain departments, budgets and vendors; view all data and reports |
| Requester | Faculty or staff member who needs an item or service | Raise purchase requests with line items; track status; resubmit when revision is required |
| HOD | Head of the requesting department; first approver | Review pending requests; record Approve, Reject or Revision Required with comments |
| Finance | Finance officer; second approver and budget owner | Review requests approved by the HOD; check budget status; record Approve or Reject with comments |

In the role hierarchy of the application the CEO role is at the top, and Finance, HOD and Requester are its sub-roles, so the CEO can see the data of all other users.

## 3.2 Functional Requirements

**Table 3.2:** Functional Requirements

| FR ID | Requirement | Role(s) | Priority |
|---|---|---|---|
| FR-01 | The system shall maintain departments with a unique department code, name, head, contact email, status and annual budget. | Admin | High |
| FR-02 | The system shall maintain a budget record for each department, financial year and budget head, with allocated, reserved and spent amounts, and shall calculate the available amount automatically. | Admin, Finance | High |
| FR-03 | The system shall maintain a vendor master with contact details, category and status. | Admin | Medium |
| FR-04 | The system shall allow a requester to raise a purchase request with priority, budget head, purpose, required-by date, department, supporting documents and one or more line items. | Requester | High |
| FR-05 | The system shall calculate the total of each line item (unit price × quantity) and the estimated total of the request automatically. | System | High |
| FR-06 | The system shall compare the estimated total with the available budget of the department and budget head and set the budget status. | System | High |
| FR-07 | The system shall let the HOD record Approve, Reject or Revision Required with comments, and shall update the approval status accordingly. | HOD | High |
| FR-08 | The system shall let Finance record Approve or Reject with comments, and shall update the approval status to Approved or Rejected by Finance. | Finance | High |

| FR ID | Requirement | Role(s) | Priority |
|---|---|---|---|
| FR-09 | The system shall write an approval-history record (stage, approver, action, date, comments, previous and new status) at every decision. | System | High |
| FR-10 | The system shall show the HOD decision field only to HOD users and the Finance decision field only to Finance users. | System | High |
| FR-11 | The system shall send an email to the requester when the HOD approves or rejects a request. | System | Medium |
| FR-12 | The system shall provide list reports for pending HOD approvals, pending Finance approvals, approved purchases, all purchases, departments, budgets, vendors and approval history. | All | Medium |
| FR-13 | The system shall provide a dashboard with request counts by status, charts and a budget overview. | Admin, HOD, Finance | Medium |
| FR-14 | The system shall allow each role to view and edit only the forms and reports permitted for that role. | All | High |

## 3.3 Non-Functional Requirements

**Table 3.3:** Non-Functional Requirements

| NFR ID | Category | Requirement |
|---|---|---|
| NFR-01 | Performance | Reports with up to 1,000 records shall load within 5 seconds. |
| NFR-02 | Security | Users shall access only the data and reports permitted by their profile; decision fields shall be hidden from other profiles. |
| NFR-03 | Usability | The application shall be usable on a desktop browser, a mobile browser and the Zoho Creator mobile app, with clear mandatory-field markers. |
| NFR-04 | Availability | The application shall be available whenever the Zoho Creator cloud service is available (hosted by Zoho). |
| NFR-05 | Maintainability | Forms, fields and workflows shall follow a clear naming convention so that another developer can change them easily. |
| NFR-06 | Auditability | Every approval decision shall be traceable through the Approval_History form, and record changes through the audit trail. |

## 3.4 Hardware and Software Requirements

**Table 3.4:** Hardware and Software Requirements

| Category | Specification |
|---|---|
| Platform | Zoho Creator [Trial / Education / Professional] Edition |
| Scripting language | Deluge (Data Enriched Language for the Universal Grid Environment) |
| Browser | Google Chrome / Microsoft Edge / Mozilla Firefox (latest version) |
| Other Zoho services | Zoho Mail (through the sendmail task of Deluge) |
| Mobile | Mobile browser and the Zoho Creator mobile app |

| Category | Specification |
|---|---|
| Client hardware | Intel Core i3/i5 or above, 8 GB RAM, internet connection (or any smartphone with internet) |

## 3.5 Use Case Model

The use case diagram shows the four user roles and the main actions each role performs in the application.

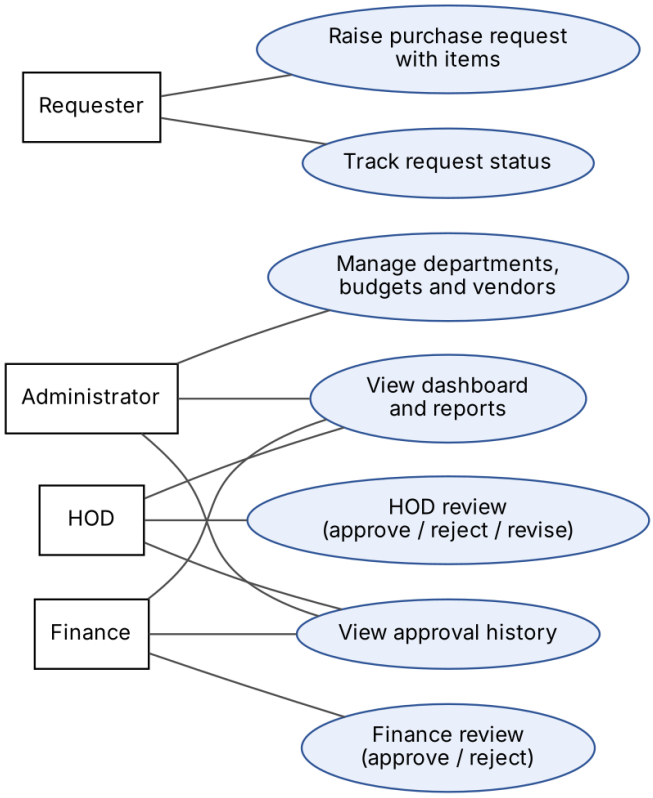



**Figure 3.1:** Use Case Diagram of the Application

**Table 3.5:** Use Case Description – Raise and Approve Purchase Request

| Item | Description |
|---|---|
| Use Case ID | UC-01 |
| Use Case Name | Raise and Approve Purchase Request |
| Actor(s) | Requester, HOD, Finance |
| Pre-condition | The department and its budget record for the financial year and budget head already exist in the system. |

| Item | Description |
|---|---|
| Main Flow | 1. The requester opens the Purchase Request form. 2. The requester selects the department, budget head and priority, enters the purpose and required-by date, and adds the items (name, category, quantity, unit price). 3. The workflow calculates the estimated total (DS-01). 4. On submit, the budget-check workflow sets the Budget_Status (DS-02). 5. The request appears in the Pending HOD Approval report. 6. The HOD records a decision (DS-03); an approved request moves to Pending Finance Approval. 7. Finance records a decision (DS-04); an approved request becomes Approved. 8. Every decision is saved in Approval_History. |
| Alternate Flow | If the HOD selects Revision Required, the request goes back to the requester for correction. If the HOD or Finance rejects, the status becomes Rejected by HOD or Rejected by Finance and the reason is kept in the comments and in the history. |
| Post-condition | The request carries its final status and the approval history contains one record per decision. |
| Related FR | FR-04 to FR-09 |

# CHAPTER 4
# SYSTEM DESIGN

## 4.1 System Architecture

The solution has four layers. The **user layer** contains the four roles, who reach the application through a desktop browser, a mobile browser or the Zoho Creator mobile app. The **application layer** is the College application on Zoho Creator, made of forms, reports, a dashboard page, Deluge scripts, form workflows, an approval process and an approval blueprint. The **data layer** is the Zoho Creator cloud database, where each form stores its records and lookup fields link the records. The **integration layer** connects the application to Zoho Mail for notifications; the Creator REST API is available for future integration with other systems.

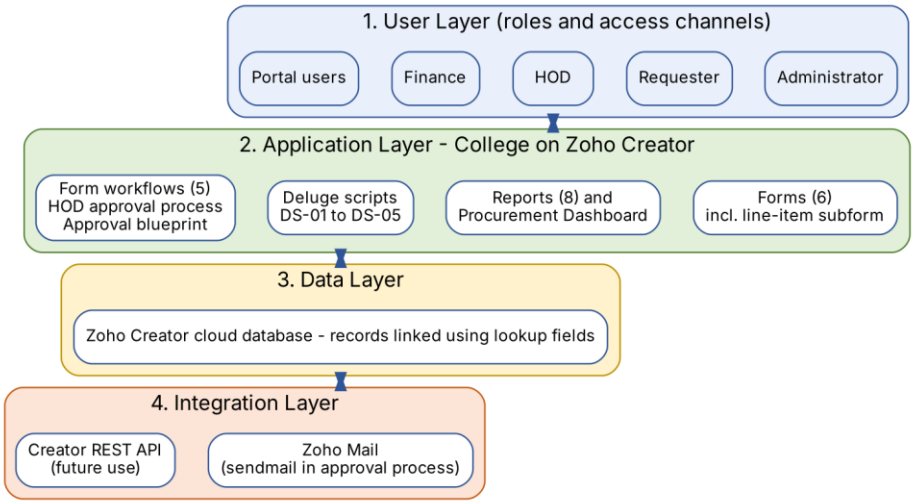



**Figure 4.1:** System Architecture of the Application

## 4.2 Data Model

The data model follows the real life cycle of a purchase. One department has many budget records and raises many purchase requests. Each purchase request has many line items and many approval-history records. The vendor form is a master list that is kept for future purchase-order work. In Zoho Creator the relationships are created with lookup fields, and the line items are held in a subform (grid) of the Purchase_Request form. The ER diagram and the table of entities are given below.

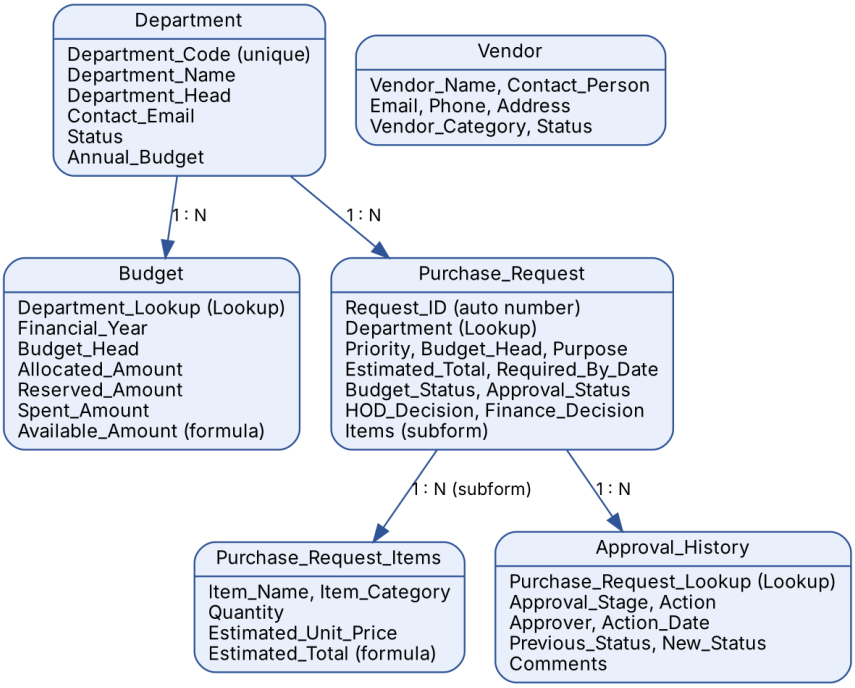



**Figure 4.2:** Entity–Relationship Diagram

**Table 4.1:** Entities and Relationships

| Entity (Form) | Key Fields | Related To | Relationship Type | Implemented Using |
|---|---|---|---|---|
| Department | Department_Code, Department_Name, Department_Head, Status, Annual_Budget | Budget, Purchase_Request | One-to-Many | Lookup Department_Lookup in Budget; lookup Department in Purchase_Request |
| Budget | Department_Lookup, Financial_Year, Budget_Head, Allocated_Amount, Reserved_Amount, Spent_Amount, Available_Amount | Department | Many-to-One | Lookup Department_Lookup |
| Vendor | Vendor_Name, Contact_Person, Email, Phone, Vendor_Category, Status | — | Stand-alone master | — |
| Purchase_Request | Request_ID, Priority, Budget_Head, Purpose, Estimated_Total, Required_By_Date, Approval_Status, HOD_Decision, Finance_Decision | Department, Purchase_Request_Items, Approval_History | Many-to-One; One-to-Many | Lookup Department; subform Items |
| Purchase_Request_Items | Item_Name, Item_Category, Quantity, Estimated_Unit_Price, Estimated_Total | Purchase_Request | Many-to-One | Subform (grid) Items |
| Approval_History | Approval_Stage, Action, Approver, Action_Date, Previous_Status, New_Status, Comments | Purchase_Request | Many-to-One | Lookup Purchase_Request_Lookup |

## 4.3 Form Design

One field specification table is given for each form. Field types are named as they appear in Zoho Creator. A red star on the form means the field is mandatory.

**Table 4.2:** Field Specification – Department Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Department Code | Department_Code | Single Line | Yes | No duplicate values |
| Department Name | Department_Name | Single Line | Yes | Name of the department |
| Department Head | Department_Head | Single Line | Yes | Name of the HOD |
| Contact Email | Contact_Email | Email | No | Valid email format; maximum 80 characters |
| Status | Status | Drop Down | Yes | Active / Inactive; default Active |
| Annual Budget | Annual_Budget | Currency (USD) | Yes | Total yearly budget of the department |

**Table 4.3:** Field Specification – Budget Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Department | Department_Lookup | Lookup | Yes | Linked to Department (shows Department_Name) |
| Financial Year | Financial_Year | Drop Down | No | 2024-2025 to 2027-2028 |
| Budget Head | Budget_Head | Drop Down | No | Equipment / Stationery / Software / Maintenance / Other |
| Allocated Amount | Allocated_Amount | Currency (INR) | Yes | Amount allocated for the head |
| Reserved Amount | Reserved_Amount | Currency (INR) | No | Amount blocked for requests in process |
| Spent Amount | Spent_Amount | Currency (INR) | No | Amount already spent |
| Available Amount | Available_Amount | Formula | No | Allocated – Reserved – Spent (null treated as 0) |

**Table 4.4:** Field Specification – Vendor Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Vendor Name | Vendor_Name | Single Line | Yes | Name of the supplier |
| Contact Person | Contact_Person | Single Line | No | Person to contact |
| Email | Email | Email | No | Valid email; maximum 80 characters |
| Phone | Phone | Phone | No | Formatted number |

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Address | Address | Multi Line | No | Postal address |
| Vendor Category | Vendor_Category | Drop Down | No | Equipment / Stationery / Software / Maintenance / Other |
| Status | Status | Drop Down | No | Active / Inactive; default Active |

**Table 4.5:** Field Specification – Purchase_Request Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Request ID | Request_ID | Auto Number | — | Starts at 1; generated by the system |
| Priority | Priority | Drop Down | No | Low / Medium / High / Urgent; default Medium |
| Budget Status | Budget_Status | Drop Down | No | Not Checked / Sufficient / Insufficient; default Not Checked |
| Budget Head | Budget_Head | Drop Down | No | Equipment / Stationery / Software / Maintenance |
| Purpose | Purpose | Multi Line | Yes | Reason for the purchase |
| Estimated Total | Estimated_Total | Currency (USD) | No | Calculated by workflow DS-01 from the items |
| Required By Date | Required_By_Date | Date | Yes | Format dd-MMM-yyyy |
| Requested By | Requested_By | Single Line | No | Name of the requester |
| Approval Status | Approval_Status | Drop Down | No | Draft / Pending HOD Approval / Rejected by HOD / Revision Required / Pending Finance Approval / Rejected by Finance / Approved / Completed; default Draft |
| Department | Department | Lookup | Yes | Linked to Department |
| Finance Comments | Finance_Comments | Multi Line | No | Entered by Finance |
| HOD Comments | HOD_Comments | Multi Line | No | Entered by HOD |
| HOD Decision | HOD_Decision | Drop Down | No | Approve / Reject / Revision Required; hidden for non-HOD users |
| Finance Decision | Finance_Decision | Drop Down | No | Pending / Approve / Reject; hidden for non-Finance users |
| Items | Items | Subform (grid) | — | Line items from Purchase_Request_Items |
| Supporting Documents | Supporting_Documents | File Upload | No | Quotations, specifications |
| Requester Email | Requester_Email | Email | No | Used for notifications |

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Request Date | Request_Date | Date | No | Date of raising the request |

**Table 4.6:** Field Specification – Purchase_Request_Items Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Item Name | Item_Name | Single Line | Yes | Name of the item |
| Item Category | Item_Category | Drop Down | No | Equipment / Stationery / Software / Maintenance / Other |
| Description | Description | Multi Line | No | Specification of the item |
| Quantity | Quantity | Number | Yes | Number of units |
| Estimated Unit Price | Estimated_Unit_Price | Currency (USD) | Yes | Price of one unit |
| Estimated Total | Estimated_Total | Formula (Decimal) | No | Estimated_Unit_Price × Quantity |

**Table 4.7:** Field Specification – Approval_History Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default / Remarks |
|---|---|---|---|---|
| Purchase Request | Purchase_Request_Lookup | Lookup | Yes | Linked to Purchase_Request (shows ID) |
| Approval Stage | Approval_Stage | Drop Down | No | HOD Review / Finance Review / Procurement |
| Action | Action | Drop Down | No | Submitted / Approved / Rejected / Revision Requested (filled with the decision) |
| Approver | Approver | Single Line | No | Logged-in user at the time of decision |
| Action Date | Action_Date | Date-Time | No | Time of the decision |
| Previous Status | Previous_Status | Single Line | No | Approval_Status before the decision |
| New Status | New_Status | Single Line | No | Approval_Status after the decision |
| Comments | Comments | Multi Line | No | HOD or Finance comments |

## 4.4 Process and Workflow Design

The flow below shows the main process: a request is raised, its total is calculated and its budget is checked, then it is reviewed by the HOD and by Finance. Every decision is written to the approval history. The calculation runs when the item quantity changes (DS-01). The budget check, the HOD decision update and the Finance decision update run in the On Success event of the Purchase_Request form (DS-02, DS-03 and DS-04).

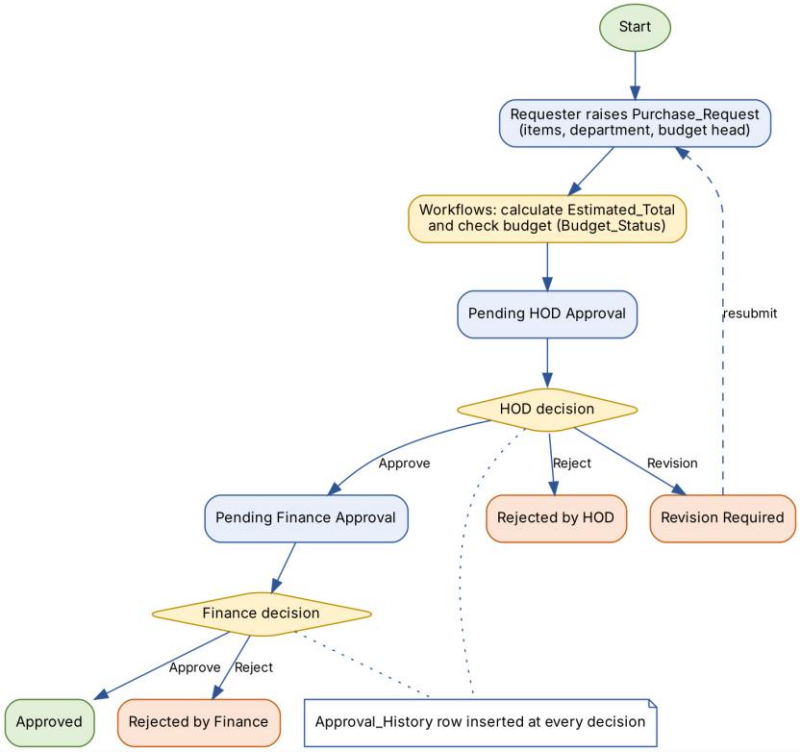



**Figure 4.3:** Workflow / Process Flow for Purchase Request Approval

## 4.5 User Interface and Navigation Design

The application menu is divided into sections that follow the business process: Approvals (pending and completed approval tasks), Department, Budget, Vendor, Purchase Request, Purchase Request Items, Approval History and the Procurement Dashboard. Under Purchase Request the user finds the form and the four reports Purchase report, Pending HOD Approval, Pending Finance Approvals and Approved Purchases. The navigation structure is shown below, and the actual screens are shown in Chapter 5.

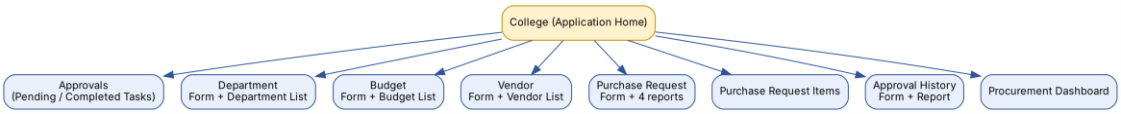



**Figure 4.4:** Navigation Structure of the College Application

## 4.6 Access Control Design

The permission matrix below is implemented in Section 5.8. A = Add (create), E = Edit, V = View, D = Delete, — = No access. The Administrator profile has all permissions. For the other profiles the matrix follows the report permissions defined in the application; every profile can open the Procurement Dashboard.

**Table 4.8:** Role–Permission Matrix

| Component | Administrator | HOD | Finance | Requester |
|---|---|---|---|---|
| Forms (create new record) | A E V D | A | A | A |
| Report: Department_List | A E V D | V | V | V |
| Report: Budget_List | A E V D | V | V | V |
| Report: Vendor_List | A E V D | V | V | V |

| Component | Administrator | HOD | Finance | Requester |
|---|---|---|---|---|
| Report: Purchase_report | A E V D | V | V | V E |
| Report: Pending_HOD_Approval | A E V D | V E | V | — |
| Report: Pending_Finance_Approvals | A E V D | V | V E | — |
| Report: Approved_Purchases | A E V D | V | V | V |
| Report: Approval_History_Report | A E V D | V | V | V |
| Page: Procurement_Dashboard | V | V | V | V |

# CHAPTER 5
# IMPLEMENTATION

## 5.1 Application Setup

The application was created in the Zoho Creator workspace with the name **College** (application icon "COL"). The forms were created in the order of the business process: Department, Budget, Vendor, Purchase_Request, Purchase_Request_Items and Approval_History. The application menu (sections) was arranged in the same order so that users follow the natural flow of a purchase. The date format is dd-MMM-yyyy, the time zone is Asia/Kolkata and and the Lato font is used. Figure 5.1 shows the Procurement Dashboard together with the left-hand navigation menu with the sections Approvals, Department, Budget, Vendor, Purchase Request, Purchase Request Items, Approval History and Procurement Dashboard.

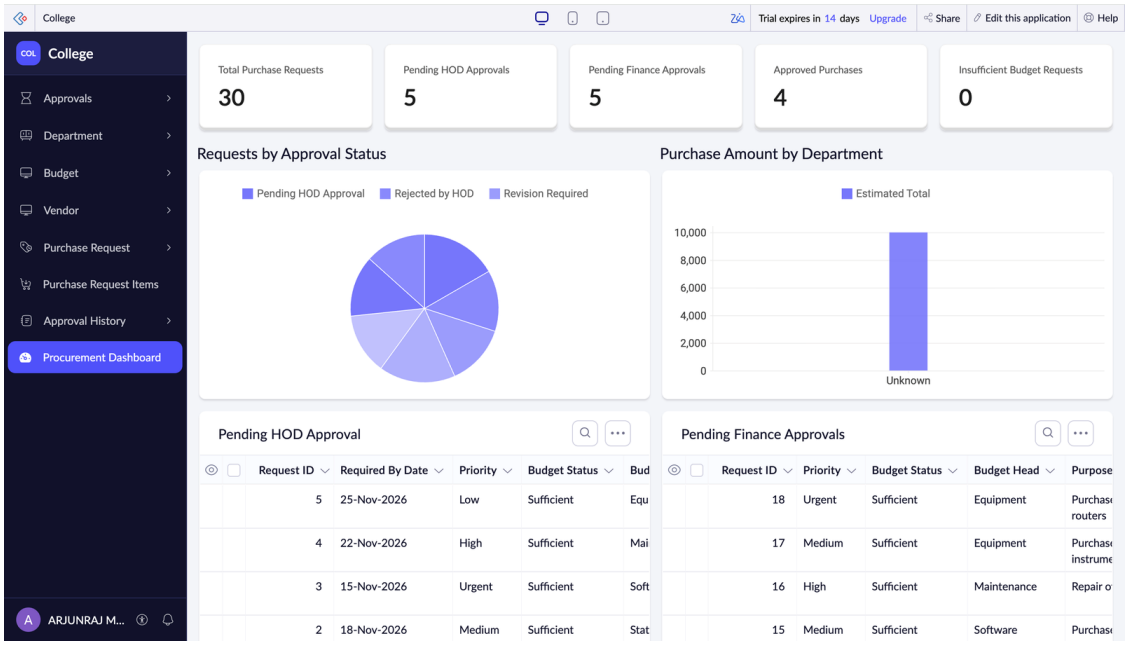



**Figure 5.1:** Procurement Dashboard and Application Navigation

## 5.2 Forms and Data Capture

Every form uses a single-column layout with the label placed on the left. The forms use the following field types: Single Line, Multi Line, Drop Down, Date, Date-Time, Currency (USD and INR), Number, Decimal, Auto Number, Email, Phone, File Upload, Lookup, Formula and Subform (grid) — fourteen types in all. Lookup fields (for example Department and Purchase_Request_Lookup) show a drop-down list of existing records, so users select a record instead of typing it again. Mandatory fields are marked with a red star, and date fields show the format dd-MMM-yyyy as a hint. The Department_Code field has the no-duplicate setting.

**Department** stores the department code, name, head, contact email, status and annual budget. **Budget** stores the allocated, reserved and spent amounts for each department, financial year and budget head, and a formula field shows the available amount. **Vendor** stores supplier details. **Purchase_Request** is the main form. It records the priority, budget status, budget head, purpose, estimated total, required-by date, requester, approval status, department, comments, decisions, supporting documents and the Items subform. In the live

form (Figure 5.2) the defaults Priority = Medium, Budget Status = Not Checked and Approval Status = Draft are visible, and Purpose, Required By Date, Department and the item fields Estimated Unit Price, Item Name and Quantity carry the red mandatory star. **Approval_History** is filled by the workflows and keeps the audit record of every decision.

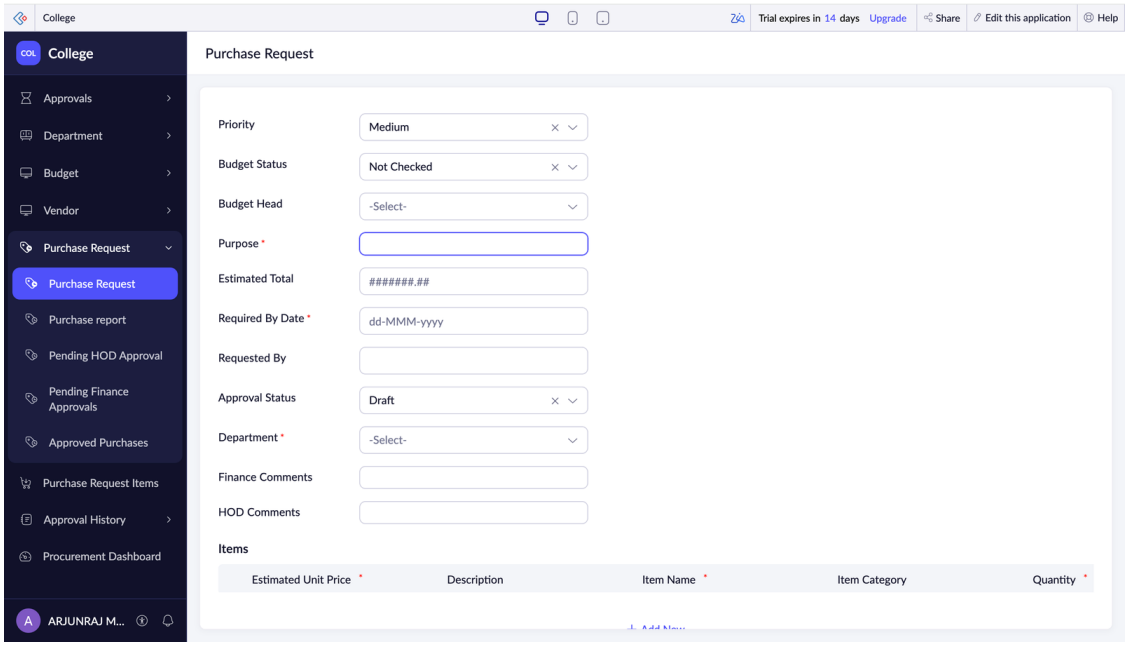



**Figure 5.2:** Purchase Request Form in Live Mode

The Decision_Field_Access workflow runs when the form loads and hides the HOD_Decision field from everyone who is not in the HOD profile and the Finance_Decision field from everyone who is not in the Finance profile, so each approver sees only his or her own decision field.

## 5.3 Reports

Eight list reports were created for the application. Three of them are filtered work-lists on the Purchase_Request form (pending HOD, pending Finance and approved), and the remaining five show the full data of a form. List reports support search, sorting, grouping and filters, and each record can be opened in the detail view. The Department_List and Budget_List reports also show total, average, minimum and maximum for the amount columns.

**Table 5.1:** Reports Implemented

| Report Name | Type | Source Form | Filter / Grouping / Sorting | Roles with Edit |
|---|---|---|---|---|
| Department_List | List | Department | Sort by Department_Name; totals on Annual_Budget | Admin |
| Budget_List | List | Budget | Sort by Department and Financial_Year; totals on amounts | Admin |
| Vendor_List | List | Vendor | Sort by Vendor_Name | Admin |
| Purchase_report | List | Purchase_Request | All requests with item total | Admin, Requester |

| Report Name | Type | Source Form | Filter / Grouping / Sorting | Roles with Edit |
|---|---|---|---|---|
| Pending_HOD_Approval | List (filtered) | Purchase_Request | Approval_Status == “Pending HOD Approval” | Admin, HOD |
| Pending_Finance_Approvals | List (filtered) | Purchase_Request | Approval_Status == “Pending Finance Approval” | Admin, Finance |
| Approved_Purchases | List (filtered) | Purchase_Request | Approval_Status == “Approved” | Admin |
| Approval_History_Report | List | Approval_History | All decisions with stage, approver and status change | Admin |

Figure 5.3 shows the Pending HOD Approval report with five requests (Request IDs 1 to 5) that are waiting for the HOD, together with their priority, budget status, budget head, purpose, estimated total and requester. Figure 5.4 shows the Approved Purchases report with four approved requests (Request IDs 23 to 26).

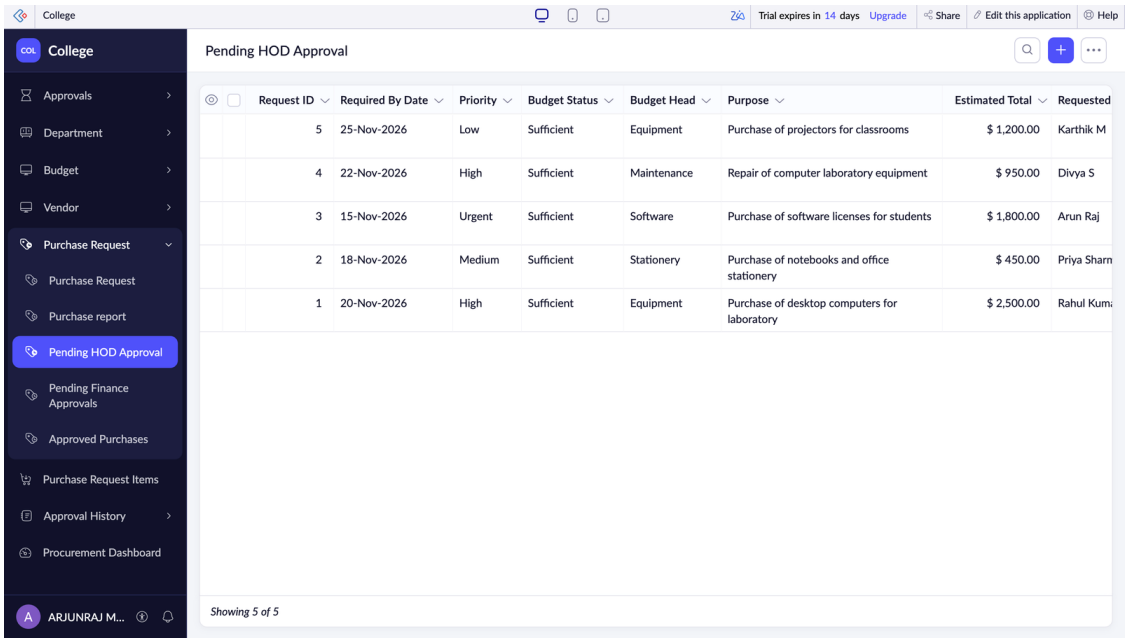



**Figure 5.3:** Pending HOD Approval Report

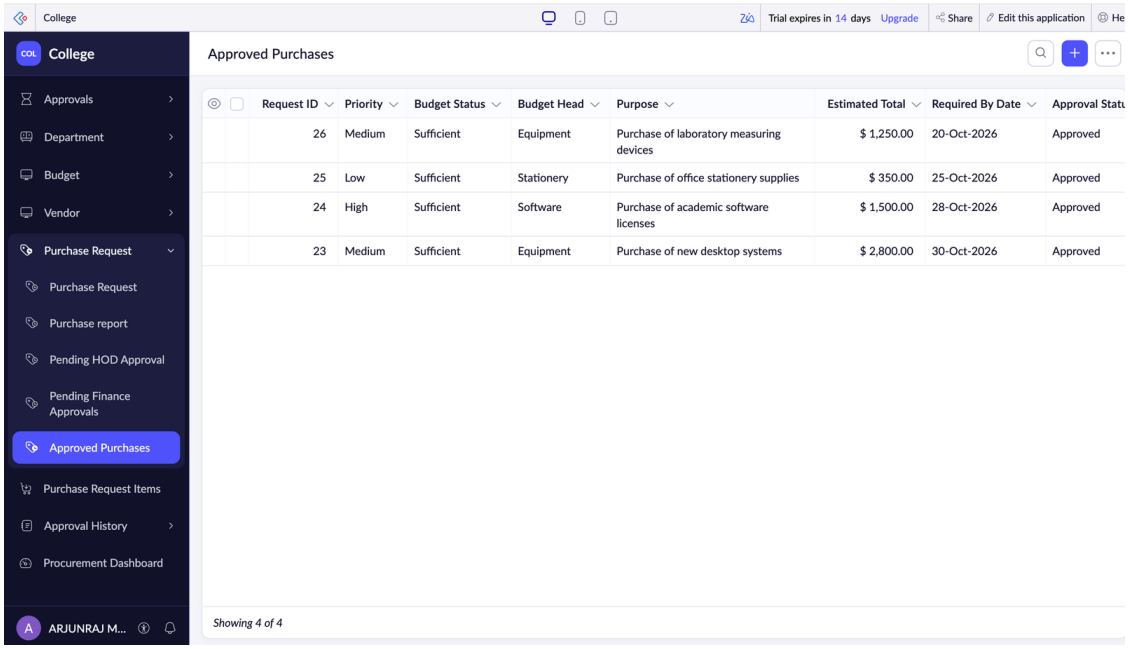



**Figure 5.4:** Approved Purchases Report

## 5.4 Pages and Dashboards

The **Procurement_Dashboard** page (Figure 5.1) was built with the Zoho Creator UI Builder for all four profiles. It contains five KPI boxes — Total Purchase Requests, Pending HOD Approvals, Pending Finance Approvals, Approved Purchases and Insufficient Budget Requests — each calculated from the Purchase_Request form with a criteria such as Approval_Status == "Pending HOD Approval". Below the KPI boxes are a pie chart "Requests by Approval Status" (count of requests by status) and a column chart "Purchase Amount by Department" (sum of Estimated_Total by department). The Pending HOD Approval and Pending Finance Approvals reports are embedded side by side without add, edit or delete buttons, so the approver can start work directly from the dashboard. Further down the page are a budget overview strip (total allocated, reserved, spent and available amounts from the Budget form) and the embedded Approval History report. In the live application the KPI boxes show 30 total requests, 5 pending HOD approvals, 5 pending Finance approvals, 4 approved purchases and 0 insufficient-budget requests.

## 5.5 Deluge Scripting

Deluge is the scripting language of Zoho Creator. Five scripts were written for the form workflows of this application. They run on form events (On Load, On User Input and On Success). The short list is given in the table, and the full code is placed in Appendix C.

**Table 5.2:** Deluge Scripts Implemented

| Script ID | Location / Trigger | Purpose | Deluge Concepts Used |
|---|---|---|---|
| DS-01 | Purchase_Request → On User Input of Items.Quantity (workflow "calculate") | Add up the item totals into Estimated_Total | for each loop, null check, arithmetic |
| DS-02 | Purchase_Request → On Success (workflow "budget-check") | Compare Estimated_Total with the available budget and set Budget_Status | criteria fetch, count(), ifnull(), if-else |

| Script ID | Location / Trigger | Purpose | Deluge Concepts Used |
|---|---|---|---|
| DS-03 | Purchase_Request → On Success (workflow "Hod_Approval") | Set Approval_Status from HOD_Decision and write the approval history | if-else, insert into, zoho.loginuser |
| DS-04 | Purchase_Request → On Success (workflow "Finance Approval Status Update") | Set Approval_Status from Finance_Decision and write the approval history | if-else, insert into, zoho.currenttime |
| DS-05 | Purchase_Request → On Load (workflow "Decision Field Access") | Hide HOD_Decision and Finance_Decision from other profiles | isUserInProfile(), hide field |

***Code 5.1: Total calculation for Purchase_Request (DS-01)***

```
total = 0;
for each item in input.Items
{
    if(item.Estimated_Total != null)
    {
        total = total + item.Estimated_Total;
    }
}
input.Estimated_Total = total;
```

DS-01 loops over the rows of the Items subform and adds the line totals (each line total is the unit price multiplied by the quantity, calculated by a formula field). The sum is placed in Estimated_Total, so the requester never calculates the amount by hand.

***Code 5.2: Budget check for Purchase_Request (DS-02)***

```
input.Budget_Status = "Not Checked";
if(input.Department != null && input.Budget_Head != null &&
input.Estimated_Total != null)
{
    budgetRecord = Budget[Department_Lookup == input.Department && Budget_Head
== input.Budget_Head];
    if(budgetRecord.count() > 0)
    {
        available = ifnull(budgetRecord.Allocated_Amount,0) -
ifnull(budgetRecord.Reserved_Amount,0) - ifnull(budgetRecord.Spent_Amount,0);
        if(input.Estimated_Total <= available)
        {
            input.Budget_Status = "Sufficient Budget";
        }
        else
        {
            input.Budget_Status = "Insufficient Budget";
        }
    }
}
```

DS-02 starts with the status Not Checked, finds the budget record of the same department and budget head, calculates the available amount (allocated minus reserved minus spent) and sets the budget status by comparing it with the estimated total.

***Code 5.3: HOD decision update (DS-03)***

```
oldStatus = input.Approval_Status;
if(input.HOD_Decision == "Approve")
```

```
{
    input.Approval_Status = "Pending Finance Approval";
}
else if(input.HOD_Decision == "Reject")
{
    input.Approval_Status = "Rejected by HOD";
}
else if(input.HOD_Decision == "Revision Required")
{
    input.Approval_Status = "Revision Required";
}
if(input.HOD_Decision == "Approve" || input.HOD_Decision == "Reject" ||
input.HOD_Decision == "Revision Required")
{
    insert into Approval_History
    [
        Purchase_Request_Lookup=input.ID
        Approval_Stage="HOD Review"
        Approver=zoho.loginuser
        Action=input.HOD_Decision
        Action_Date=zoho.currenttime
        Comments=input.HOD_Comments
        Previous_Status=oldStatus
        New_Status=input.Approval_Status
        Added_User=zoho.loginuser
    ]
}
```

DS-03 keeps the old status, moves the request to the next status according to the HOD decision, and inserts an Approval_History record with the stage, approver, action, time, comments and the old and new status. DS-04 does the same for the Finance decision (Approve gives Approved, Reject gives Rejected by Finance). DS-05 hides the decision fields from users who are not in the matching profile. The code of DS-04 and DS-05 is given in Appendix C.

## 5.6 Workflows and Automation

Workflows join the forms into one automatic process. Five form workflows run the scripts above, one approval process handles the HOD review with email notification, and one approval blueprint defines the stages of the approval history.

**Table 5.3:** Workflows Implemented

| Workflow Name | Type | Trigger / Event | Action(s) | Script Ref. |
|---|---|---|---|---|
| calculate | Form Workflow | Purchase_Request: add or edit; On User Input of Items.Quantity | Calculate Estimated_Total from the items | DS-01 |
| budget-check | Form Workflow | Purchase_Request: add or edit; On Success | Set Budget_Status | DS-02 |
| Hod_Approval | Form Workflow | Purchase_Request: add or edit; On Success | Update Approval_Status; insert Approval_History | DS-03 |
| Finance Approval Status Update | Form Workflow | Purchase_Request: add or edit; On Success | Update Approval_Status; insert Approval_History | DS-04 |
| Decision Field Access | Form Workflow | Purchase_Request: add or edit; On Load | Hide decision fields by profile | DS-05 |

| Workflow Name | Type | Trigger / Event | Action(s) | Script Ref. |
|---|---|---|---|---|
| HOD Approval | Approval | Purchase_Request, level 1 | On approve: email the submitter, set HOD_Decision = Approve and status Pending Finance Approval. On reject: email the submitter, set HOD_Decision = Reject and status Rejected by HOD | — |
| Approval Workflow | Blueprint | Approval_History; start stage Submitted | 7 stages and 11 transitions (see Figure 5.5) | — |

The blueprint "Approval Workflow" has the stages Submitted, HOD Review, Finance Review, Procurement, Approved, Revision Requested and Rejected. Its transitions are: Submit for HOD Review; Approve to Finance, Reject and Request Revision (from HOD Review); Approve to Procurement, Reject and Request Revision (from Finance Review); Final Approval, Reject and Request Revision (from Procurement); and Resubmit (from Revision Requested back to Submitted). A second, empty blueprint draft named "Approval blueprint" also exists in the application and can be removed. Schedule and batch workflows were not used because the purchase cycle is event-driven and no periodic job is needed.

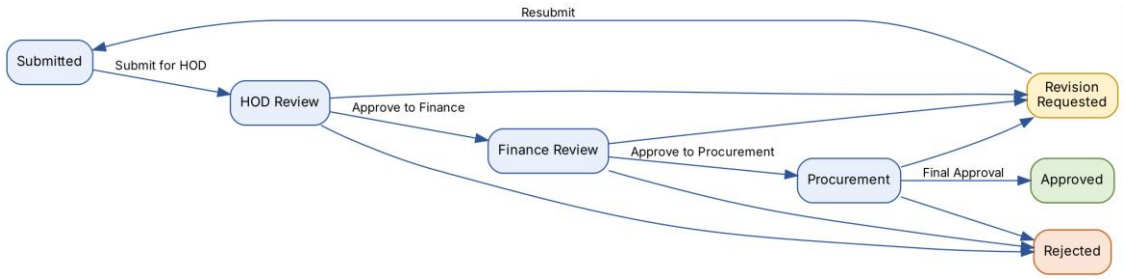



**Figure 5.5:** Approval Blueprint – Stages and Transitions

## 5.7 Integrations

The application is integrated with **Zoho Mail** through the built-in sendmail task of Deluge inside the HOD Approval process. The sender is the logged-in approver and the receiver is the submitter of the request. No password or API key is stored in any script. One email is sent when the HOD approves a request ("Your request has been approved.") and one when the HOD rejects it ("Your request has been rejected.").

**Table 5.4:** Integrations Implemented

| Integration | Service / Endpoint | Method | Authentication | Data Exchanged / Purpose |
|---|---|---|---|---|
| Email notification | Zoho Mail | Deluge sendmail task (approval actions) | Zoho account (built-in) | Approval and rejection mails to the requester |
| REST API (future) | Zoho Creator REST API | Creator API v2 | OAuth 2.0 | Planned for accounting or purchase-order systems; not used in this version |

The invokeurl task, webhooks and the Creator REST API were not needed in this version because all data is created and used inside the application. They are described in Section 9.2 as future work.

## 5.8 User Management and Role-Based Access

Access is controlled in four ways. First, a role hierarchy is defined: the CEO role is at the top, and Finance, HOD and Requester are its sub-roles. Second, user permission profiles define what each kind of user can do: Administrator (all permissions), HOD, Finance, Requester, Read, Write and Developer. For example, the HOD profile can view and edit the Pending HOD Approval report, the Finance profile can view and edit the Pending Finance Approvals report, and the Requester profile can view and edit the Purchase report but cannot open the two approval work-lists. Third, portal profiles are defined for users who log in through a portal: HOD, Finance Portal, Requester Portal and the default Customer profile (Figure 5.6). The portal HOD profile has view-all and modify-all rights on Purchase_Request so that the HOD can see every request of the department. Fourth, the Decision_Field_Access workflow hides the decision fields from other profiles, as described in Section 5.2.

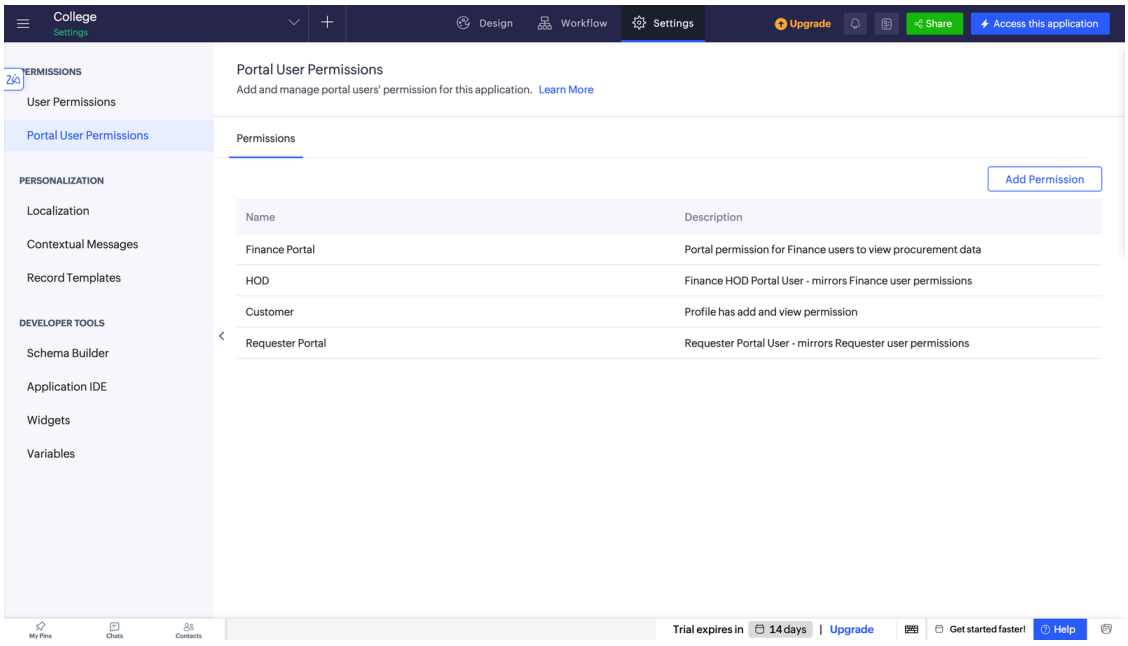



**Figure 5.6:** Portal User Permissions Configuration

## 5.9 AI and Analytics Features

Analytics is provided by the pie chart, the column chart, the KPI boxes and the budget overview strip of the dashboard, which are calculated from the live data. No Zia prediction or other AI feature is used in the version submitted. The scripts and forms were written by the student; where an AI assistant was used to understand Deluge syntax or to prepare this report, the code and content were checked in the Deluge editor and in the live application before use. Zia text analysis of quotations and spend forecasting are planned as future work.

## 5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst function, JavaScript widget or custom REST service) was used, because all required logic was possible with Deluge scripts and standard

workflows, which is easier to maintain. If heavy processing is needed in future, for example reading scanned quotations or forecasting spend, a Catalyst function can be called from Deluge through the invokeurl task.

## 5.11 Mobile and Portal Access

The application has a web layout, a tablet layout and a phone layout, and the device icons at the top of the live application (visible in Figures 5.1 to 5.4) allow the developer to preview each of them. On a phone the form fields stack in one column and drop-downs open as touch-friendly lists, so an approver can review and decide from anywhere using a mobile browser or the Zoho Creator mobile app. Requesters, HODs and finance users can be given portal access through the portal profiles described in Section 5.8.

# CHAPTER 6
# TESTING AND DEBUGGING

## 6.1 Testing Strategy

Testing is planned at four levels. Unit testing checks each form (mandatory fields, field types, lookups, formulas) and each Deluge script separately. Integration testing checks that a record created in one form appears correctly in the next through lookups and that the workflows run in the right order. Role-based testing is done by logging in with each profile and trying allowed and not-allowed actions. User acceptance testing is done by asking other users to complete a full purchase request.

## 6.2 Test Cases

Table 6.1 lists the test cases. The status "Pass" is entered only for the cases that are supported by the live screens in Chapter 5. The remaining cases are to be executed in the live application, and their Actual Result and Status columns are to be filled by the student after execution.

**Table 6.1:** Test Cases

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC-01 | FR-01 | Create a department with a duplicate code | Enter an existing Department_Code; Submit | Duplicate value error shown | To execute |
| TC-02 | FR-02 | Budget available amount | Allocated 10000, Reserved 2000, Spent 3000 | Available_Amount shows 5000 | To execute |
| TC-03 | FR-04 | Submit request without mandatory fields | Leave Purpose, Required By Date and Department empty; Submit | Mandatory field errors shown (red star fields) | To execute |
| TC-04 | FR-05 | Estimated total from items | Add 2 items: 5 × 100 and 2 × 250 | Estimated_Total = 1000 | To execute |
| TC-05 | FR-06 | Budget sufficient | Total less than available amount | Budget_Status = Sufficient Budget | To execute |
| TC-06 | FR-06 | Budget insufficient | Total more than available amount | Budget_Status = Insufficient Budget | To execute |
| TC-07 | FR-07 | HOD approves | HOD_Decision = Approve | Status = Pending Finance Approval; history row added | To execute |
| TC-08 | FR-07 | HOD rejects | HOD_Decision = Reject | Status = Rejected by HOD; history row added | To execute |
| TC-09 | FR-07 | HOD asks for revision | HOD_Decision = Revision Required | Status = Revision Required; history row added | To execute |
| TC-10 | FR-08 | Finance approves | Finance_Decision = Approve | Status = Approved; history row added | To execute |
| TC-11 | FR-08 | Finance rejects | Finance_Decision = Reject | Status = Rejected by Finance; history row added | To execute |
| TC-12 | FR-11 | Email on HOD approval | Approve through the HOD Approval process | Requester receives the approval email | To execute |

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC-13 | FR-12 | Pending HOD report | Open Pending HOD Approval | Only requests with status Pending HOD Approval are listed (5 of 5 shown) | Pass |
| TC-14 | FR-12 | Approved Purchases report | Open Approved Purchases | Only requests with status Approved are listed (4 of 4 shown) | Pass |
| TC-15 | FR-13 | Dashboard figures | Open Procurement Dashboard | KPI counts equal the report counts (5 pending HOD, 4 approved) | Pass |
| TC-16 | FR-14 | Portal profiles configured | Open Settings → Portal User Permissions | HOD, Finance Portal, Requester Portal and Customer profiles are listed | Pass |
| TC-17 | FR-10 | Decision field access | Open the form as a Requester | HOD_Decision and Finance_Decision are hidden | To execute |

## 6.3 Debugging Deluge Scripts

Scripts are debugged with the info statement, which prints values in the execution log, and with the Deluge editor's test option. Error messages shown on form submission are also read carefully. Common problems in a project of this kind are a lookup field returning an ID instead of a record, a picklist value written in a script that does not exist in the field's choice list, and null values in arithmetic. These are solved by fetching the record with the criteria ID == input.field before reading its values, by checking the exact choice text of the field, and by using ifnull() as done in the Budget form formula and in DS-02.

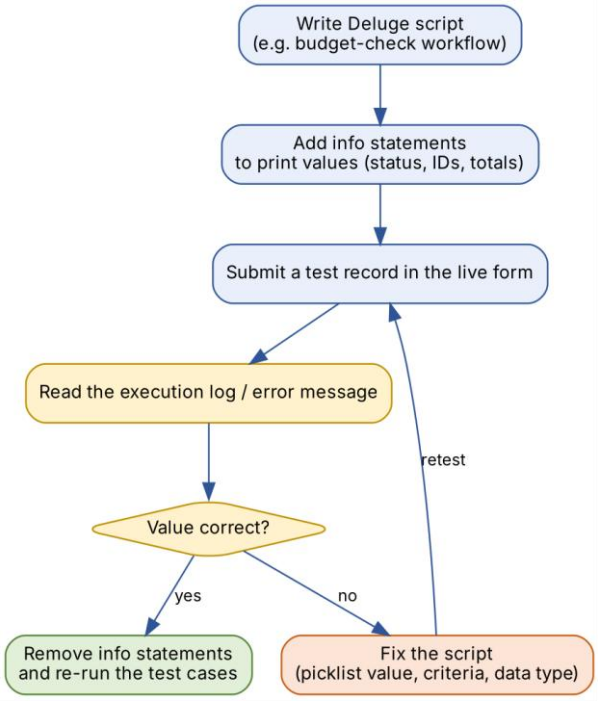



**Figure 6.1:** Deluge Debugging Process Using info Statements

## 6.4 Defect Log

The defects below were found when the application configuration and the live screens were reviewed. They are recorded honestly as open items with the fix to be applied.

**Table 6.2:** Defect Log

| Bug ID | Description | Root Cause | Fix to Apply | Status |
|---|---|---|---|---|
| BUG-01 | The KPI box “Insufficient Budget Requests” always shows 0 and the budget-check script writes values that are not in the Budget_Status drop-down. | The drop-down choices are Sufficient and Insufficient, but DS-02 writes “Sufficient Budget” and “Insufficient Budget”, and the KPI criteria uses “Insufficient Budget”. | Use one set of values in the field, in DS-02 and in the KPI criteria. | Open |
| BUG-02 | The chart “Purchase Amount by Department” shows the whole amount under “Unknown”. | The Department lookup of the sample requests is empty or is displayed by ID instead of by name. | Fill the department for every request and use Department_Name as the display value. | Open |
| BUG-03 | Approval_History.Action may not accept the decision text written by DS-03 and DS-04. | The Action drop-down has Submitted / Approved / Rejected / Revision Requested, but the scripts write Approve / Reject / Revision Required. | Align the choice list with the decision values or map the decision to the action text in the script. | Open |
| BUG-04 | Amounts are shown in different currencies. | Department and Purchase_Request use USD fields, while Budget uses INR fields. | Use one currency (INR) for all amount fields. | Open |
| BUG-05 | Budget_Head in Purchase_Request has no “Other” choice, but Budget and Items have it. | Choice lists were created separately. | Add “Other” to the Purchase_Request list. | Open |

## 6.5 User Acceptance Testing

User acceptance testing is carried out by asking real users of each role to complete a full request. The feedback is to be recorded below after the sessions.

**Table 6.3:** User Acceptance Feedback

| User | Role Tested | Feedback | Action Taken |
|---|---|---|---|
| User 1 | Requester | [to be filled after testing] | [to be filled] |
| User 2 | HOD | [to be filled after testing] | [to be filled] |
| User 3 | Finance | [to be filled after testing] | [to be filled] |

# CHAPTER 7
# DEPLOYMENT, SECURITY AND GOVERNANCE

## 7.1 Environments and Deployment Strategy

The application is built and tested in the **Development** environment with sample data. After the test cases pass, it is published to **Production** with version notes. In production, users, roles and connections are configured again, and the sample data is removed. If a serious error is found after release, the previous version is restored using the version history, and the fix is made again in development.

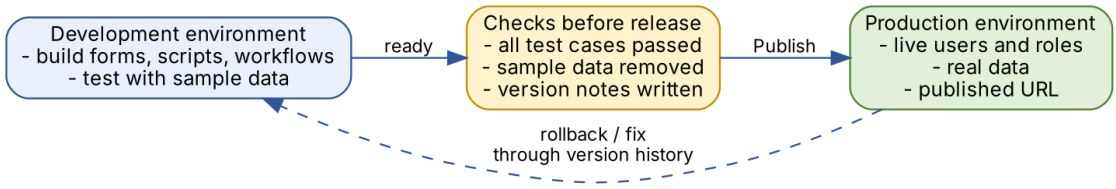



**Figure 7.1:** Environments and Publish to Production (Promotion Flow)

**Table 7.1:** Deployment Checklist

| S.No. | Checklist Item | Done (✓) |
|---|---|---|
| 1 | All test cases executed in Development | |
| 2 | Sample / test data removed or isolated | |
| 3 | Roles, profiles and users configured in Production | |
| 4 | Email sending (Zoho Mail) verified in Production | |
| 5 | Workflows, approval process and blueprint enabled | |
| 6 | Application published and URL verified | |
| 7 | Version / release notes recorded | |

## 7.2 Performance Optimisation

The following techniques are used to keep the application fast. The effect was not measured with a timer in this version; measured values can be added after load testing.

**Table 7.2:** Performance Optimisation Techniques

| Technique | Component | Benefit |
|---|---|---|
| Criteria in the fetch (Department and Budget_Head) | DS-02 | Only the matching budget record is read instead of the whole Budget form |
| Filtered reports (criteria on Approval_Status) | Pending_HOD_Approval, Pending_Finance_Approvals, Approved_Purchases | Work-lists load only the records that need action |

| Technique | Component | Benefit |
|---|---|---|
| No fetch statement inside loops | DS-01 | The loop only adds the values already present in the subform |
| Formula fields instead of scripts | Available_Amount, item Estimated_Total | Values are calculated by the platform without extra script calls |
| Limited columns in reports | Purchase_report and work-lists | Less data is loaded and displayed |

## 7.3 Security Features

Security is applied at several levels: profile permissions control which forms and reports a user can open and edit; portal profiles separate portal users from internal users; the decision fields are hidden from users who are not approvers; the role hierarchy limits who can see whose data; no password or key is written inside a script; and two-factor authentication of the Zoho account is recommended for administrators.

**Table 7.3:** Security Controls Implemented

| Security Control | Where Applied | Threat Mitigated |
|---|---|---|
| Profile permissions (Administrator, HOD, Finance, Requester, Read, Write, Developer) | All forms, reports and the dashboard | Unauthorised access to data and functions |
| Portal profiles (HOD, Finance Portal, Requester Portal, Customer) | Portal access | External users seeing more than they need |
| Field hiding on form load (DS-05) | HOD_Decision, Finance_Decision | A requester changing an approval decision |
| Role hierarchy (CEO → Finance, HOD, Requester) | Data sharing | Users seeing data of other roles |
| Built-in sendmail (no stored keys) | HOD Approval process | Credential leakage |
| Approval history | Approval_History form | Denial of who decided and when |

## 7.4 Governance and Administration

The audit trail of Zoho Creator records who added or changed a record and when, and the Approval_History form keeps a business-level record of every decision. A naming convention is followed: form names and field link names use words joined with underscores (for example Purchase_Request and Required_By_Date). Data can be exported from reports as a backup. The Administrator manages user licences and is responsible for maintenance, version notes and removing access for users who have left.

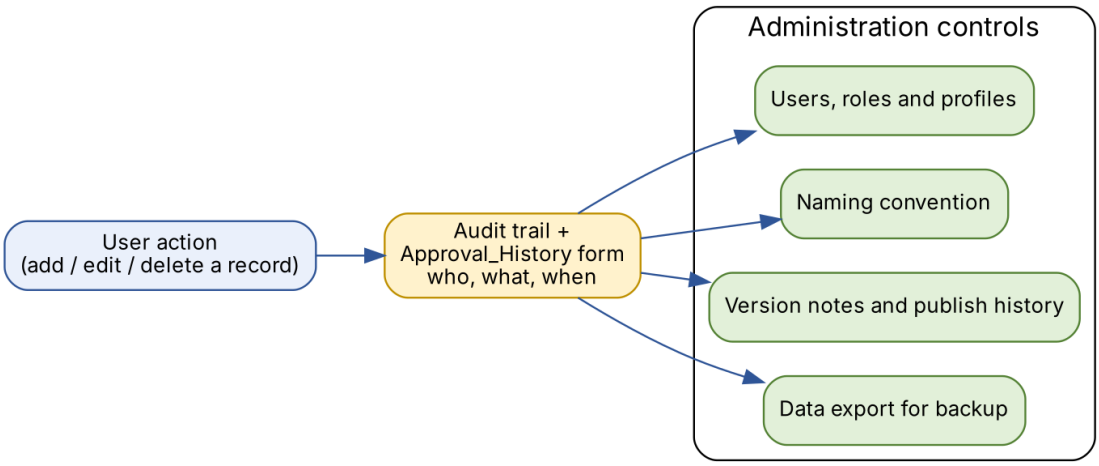



**Figure 7.2:** Audit Trail and Administration Controls

# CHAPTER 8
# RESULTS AND DISCUSSION

## 8.1 Application Output

A realistic scenario shows how the application works from end to end. The administrator creates the department and its budget for the financial year. A requester opens the Purchase Request form (Figure 5.2), selects the department and budget head, enters the purpose, required-by date and the items; the estimated total is calculated and the budget status is set. The request then appears in the Pending HOD Approval report (Figure 5.3). The HOD records a decision; an approved request moves to Pending Finance Approval, where Finance records the final decision and the request becomes Approved (Figure 5.4). Each decision is saved in Approval History, and the Procurement Dashboard (Figure 5.1) shows the updated counts.

Table 8.1 lists the sample records that are visible in the live application and that were used to check the work-lists and the dashboard.

**Table 8.1:** Sample Records Observed in the Live Application

| Request ID | Budget Head | Purpose | Estimated Total | Priority | Status |
|---|---|---|---|---|---|
| 1 | Equipment | Purchase of desktop computers for laboratory | $ 2,500.00 | High | Pending HOD Approval |
| 2 | Stationery | Purchase of notebooks and office stationery | $ 450.00 | Medium | Pending HOD Approval |
| 3 | Software | Purchase of software licenses for students | $ 1,800.00 | Urgent | Pending HOD Approval |
| 4 | Maintenance | Repair of computer laboratory equipment | $ 950.00 | High | Pending HOD Approval |
| 5 | Equipment | Purchase of projectors for classrooms | $ 1,200.00 | Low | Pending HOD Approval |
| 23 | Equipment | Purchase of new desktop systems | $ 2,800.00 | Medium | Approved |
| 24 | Software | Purchase of academic software licenses | $ 1,500.00 | High | Approved |
| 25 | Stationery | Purchase of office stationery supplies | $ 350.00 | Low | Approved |
| 26 | Equipment | Purchase of laboratory measuring devices | $ 1,250.00 | Medium | Approved |

The five pending requests add up to $ 6,900 and the four approved requests to $ 5,900. The dashboard shows 30 requests in total, of which 5 are pending HOD approval, 5 are pending Finance approval and 4 are approved; the remaining requests are in other statuses such as Rejected by HOD and Revision Required, as the pie chart shows.

## 8.2 Achievement of Objectives

**Table 8.2:** Mapping of Objectives to Outcomes

| Objective (Sec. 1.3) | Implemented Through | Outcome / Evidence |
|---|---|---|
| Design a data model and forms | Sections 4.2, 4.3, 5.2 | Six linked forms with lookups and a line-item subform; Figure 5.2 |
| Automate calculation, budget check, status change and history | Sections 5.5, 5.6 | Scripts DS-01 to DS-05 and five form workflows |
| Two-level approval flow with hidden decision fields | Sections 5.2, 5.6 | HOD Approval process, approval blueprint and Decision_Field_Access workflow |
| Role-based access, reports and dashboard | Sections 5.3, 5.4, 5.8 | Seven user profiles, four portal profiles, eight list reports and one dashboard; Figures 5.1, 5.3, 5.4, 5.6 |
| Email notification through Zoho Mail | Section 5.7 | Approve and reject mails sent by the HOD Approval process |
| Web and mobile use, secure deployment | Chapter 7, Section 5.11 | Web, tablet and phone layouts; deployment strategy in Chapter 7 |

## 8.3 Comparison with the Existing System

**Table 8.3:** Existing System versus Proposed Application

| Parameter | Existing System | Proposed Application | Improvement |
|---|---|---|---|
| Time to process one request | Days (paper form carried between offices) | Minutes for data entry; instant total and budget check | Much faster |
| Manual steps | Many (copying, calculating, signing, posting) | Few (select, add items, decide) | Less manual work |
| Budget check | Manual, in a register | Automatic by department and budget head | Fewer overspending cases |
| Approval record | Signature or message | Approval_History with approver, time and comments | Full audit trail |
| Report generation | Manual, in spreadsheets | Automatic reports and dashboard | Always up to date |
| Security | Open spreadsheets | Profiles, portal profiles and hidden decision fields | Data protected |

*The values above are qualitative estimates based on the working of the old and the new process; they should be replaced with measured timings if available.*

## 8.4 Limitations of the Low-Code Approach

Some limits were noticed while building the application. The look of the forms can be changed only within the options given by the platform. Deluge has limits on the number of statements and on API calls, so very large loops must be avoided. The application also depends on Zoho (vendor lock-in), so moving it to another platform would need rebuilding. Within this version, the Reserved and Spent amounts of the Budget form are not updated

automatically when a request is approved, the status Pending HOD Approval is not set by a script when a request is submitted, and the drop-down values of some fields need to be aligned with the scripts (Section 6.4). These items are planned as improvements.

## 8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code: forms, fields, lookups, reports, the dashboard, profiles and the approval process were configured with the mouse. The low-code part is the five Deluge scripts for calculation, budget check, status update, history logging and field hiding. A pro-code part was not needed now, but would be needed for vendor bidding analysis, accounting-system integration or reading scanned quotations. A move to pro-code is sensible when the logic becomes too complex or too slow for Deluge, or when an external system needs a special interface.

# CHAPTER 9
# CONCLUSION AND FUTURE ENHANCEMENTS

## 9.1 Conclusion

The College Procurement Management System was built on Zoho Creator to manage the purchase request cycle of a college. Six connected forms store department, budget, vendor, request, item and approval-history data. Deluge scripts and workflows calculate the request total, check the budget, update statuses and keep an approval history, while an approval process, a blueprint and role-based profiles control who can decide what. Reports and a dashboard give a clear view of all requests.

The project shows that a useful business application can be built in a short time with low-code tools. It replaces paper forms and spreadsheets with one secure online system that can be used on a computer or a mobile phone, and it can be improved easily when approval rules change.

## 9.2 Future Enhancements

- Automatic update of the Reserved and Spent amounts of the budget when a request is submitted, approved or closed.
- A workflow that sets the status to Pending HOD Approval on submission and sends email or in-app notifications at every stage, including Finance.
- Purchase-order generation for the vendor and linking of vendors to requests, with quotation comparison.
- A schedule workflow that reminds approvers of requests that cross the required-by date.
- Zia-based reading of uploaded quotations and spend analysis by department.
- Integration with an accounting system and with Zoho Analytics through the Creator REST API.

## 9.3 Learning Outcomes and Career Relevance

Through this project I learned how to design a data model, build forms with subforms and formula fields, create reports and a dashboard, write Deluge scripts, configure workflows, an approval process and a blueprint, manage profiles and portal users, test an application and prepare it for production. These skills match the work of a citizen developer, a low-code developer, a solutions consultant and an automation engineer. The project also showed me the limits of low-code and when a pro-code extension is the better choice.

# REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/

[2] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/

[3] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/

[4] A. C. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

[5] R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, “Low-Code as Enabler of Digital Transformation in Manufacturing Industry,” Applied Sciences, vol. 10, no. 1, p. 12, 2020.

[6] A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020, pp. 171–178.

# APPENDIX A
# ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = done, P = partly done, N = not used.

**Table A.1:** Feature Coverage Checklist

| Unit | Zoho Creator Feature | Done | Where in the Application | Report Section | Figure / Code No. |
|---|---|---|---|---|---|
| I | Application created & sections/menu organised | Y | Eight menu sections | 5.1 | Fig. 5.1 |
| II | Forms with ≥ 8 distinct field types | Y | All six forms (14 types) | 5.2 | Fig. 5.2 |
| II | Lookup field (relationship between forms) | Y | Department, Department_Lookup, Purchase_Request_Lookup | 4.2, 5.2 | Fig. 4.2 |
| II | Subform (parent–child data) | Y | Items grid in Purchase_Request | 4.2, 5.2 | Fig. 5.2 |
| II | Field validations / layout rules | Y | Mandatory, unique Department_Code, field hiding on load | 5.2 | Code 5.1–5.3 |
| II | List report with filter, grouping, custom action | P | Filtered list reports; no grouping or custom action | 5.3 | Fig. 5.3, 5.4 |
| II | Kanban / Calendar / Timeline / Map report | N | Not used | 5.3 | — |
| II | Pivot table / chart report | P | Pie and column charts on the dashboard page (no separate chart report) | 5.4 | Fig. 5.1 |
| II | Page / dashboard built with UI Builder | Y | Procurement_Dashboard | 5.4 | Fig. 5.1 |
| II | Deluge – form events (validate / success / field actions) | Y | On Success, On Load, On User Input | 5.5 | Code 5.1–5.3 |
| II | Deluge – functions, fetch, loops, collections | Y | Fetch in DS-02; for each in DS-01; insert in DS-03/04 | 5.5 | Code 5.1, 5.2 |
| II | Email / in-app notifications | Y | Approval process sendmail | 5.6, 5.7 | Table 5.3 |
| II | Form workflow | Y | Five form workflows | 5.6 | Table 5.3 |
| II | Scheduled workflow | N | Not used (see 5.6) | 5.6 | — |
| II | Approval process | Y | HOD Approval | 5.6 | Table 5.3 |
| II | Blueprint | Y | Approval Workflow (7 stages, 11 transitions) | 5.6 | Fig. 5.5 |
| II | Batch / report workflow | N | Not used (see 5.6) | 5.6 | — |
| II | Testing & debugging evidence | P | Test cases and defect log; execution pending | 6 | Fig. 6.1 |
| III | Integration with a Zoho service | Y | Zoho Mail (sendmail) | 5.7 | Table 5.4 |
| III | Connection / REST API call (invokeurl) | N | Not used (see 5.7) | 5.7 | — |
| III | Webhook or Creator REST API | N | Not used (see 5.7) | 5.7 | — |
| III | Users, roles and permission profiles | Y | Seven user profiles, role hierarchy | 5.8 | Fig. 5.6 |
| III | Record-level access criteria / customer portal | Y | Four portal profiles | 5.8 | Fig. 5.6 |
| III | Zia / AI feature | N | Planned (see 5.9) | 5.9 | — |

| Unit | Zoho Creator Feature | Done | Where in the Application | Report Section | Figure / Code No. |
|---|---|---|---|---|---|
| III | Analytics (charts / Zoho Analytics) | Y | Dashboard charts and KPIs | 5.9 | Fig. 5.1 |
| IV | Performance optimisation applied & measured | P | Applied; measurement pending | 7.2 | Table 7.2 |
| IV | Deployment across environments | P | Strategy defined; confirm production publish | 7.1 | Fig. 7.1 |
| IV | Security controls (permissions, hidden fields) | Y | Profiles, portal profiles, field hiding | 7.3 | Table 7.3 |
| IV | Governance (audit trail, administration) | Y | Audit trail, Approval_History, naming convention | 7.4 | Fig. 7.2 |
| V | Pro-code extension (Catalyst / widgets / custom functions) | N | Not used (see 5.10) | 5.10 | — |
| V | Low-code limitations & spectrum analysis | Y | Discussion in Chapter 8 | 8.4, 8.5 | — |
| — | Mobile app / portal access | Y | Phone and tablet layouts; portal profiles | 5.11 | Fig. 5.6 |

**Total features implemented: 20 fully and 5 partly out of 32.**

# APPENDIX B
# COURSE OUTCOME MAPPING

**Table A.2:** Mapping of Course Outcomes to Report Chapters

| CO | Course Outcome Statement | Evidenced In | Self-Assessed Level | Examiner Level |
|---|---|---|---|---|
| CO1 | Explain the principles, architecture, and business value of low-code application development using Zoho Creator. | Ch. 1, 2 | | |
| CO2 | Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting. | Ch. 4, 5 (5.1–5.6), 6 | | |
| CO3 | Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security. | Ch. 4.6, 5.7–5.9 | | |
| CO4 | Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices. | Ch. 7 | | |
| CO5 | Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer. | Ch. 5.9, 5.10, 8.4, 8.5, 9.3 | | |

# APPENDIX C
# COMPLETE DELUGE SOURCE CODE

## C.1 Purchase_Request – Calculate (DS-01)

***// DS-01 | Form: Purchase_Request | Event: On User Input of Items.Quantity***

```
total = 0;
for each item in input.Items
{
    if(item.Estimated_Total != null)
    {
        total = total + item.Estimated_Total;
    }
}
input.Estimated_Total = total;
```

## C.2 Purchase_Request – Budget Check (DS-02)

***// DS-02 | Form: Purchase_Request | Event: On Success***

```
input.Budget_Status = "Not Checked";
if(input.Department != null && input.Budget_Head != null &&
input.Estimated_Total != null)
{
    budgetRecord = Budget[Department_Lookup == input.Department && Budget_Head
== input.Budget_Head];
    if(budgetRecord.count() > 0)
    {
        available = ifnull(budgetRecord.Allocated_Amount,0) -
ifnull(budgetRecord.Reserved_Amount,0) - ifnull(budgetRecord.Spent_Amount,0);
        if(input.Estimated_Total <= available)
        {
            input.Budget_Status = "Sufficient Budget";
        }
        else
        {
            input.Budget_Status = "Insufficient Budget";
        }
    }
}
```

## C.3 Purchase_Request – HOD Decision (DS-03)

***// DS-03 | Form: Purchase_Request | Event: On Success***

```
oldStatus = input.Approval_Status;
if(input.HOD_Decision == "Approve")
{
    input.Approval_Status = "Pending Finance Approval";
}
else if(input.HOD_Decision == "Reject")
{
    input.Approval_Status = "Rejected by HOD";
}
else if(input.HOD_Decision == "Revision Required")
{
    input.Approval_Status = "Revision Required";
}
if(input.HOD_Decision == "Approve" || input.HOD_Decision == "Reject" ||
input.HOD_Decision == "Revision Required")
{
    historyRecord = Approval_History[ID == 0];
    insert into Approval_History
```

```
    [
        Purchase_Request_Lookup=input.ID
        Approval_Stage="HOD Review"
        Approver=zoho.loginuser
        Action=input.HOD_Decision
        Action_Date=zoho.currenttime
        Comments=input.HOD_Comments
        Previous_Status=oldStatus
        New_Status=input.Approval_Status
        Added_User=zoho.loginuser
    ]
}
```

## C.4 Purchase_Request – Finance Decision (DS-04)

***// DS-04 | Form: Purchase_Request | Event: On Success***

```
oldStatus = input.Approval_Status;
if(input.Finance_Decision == "Approve")
{
    input.Approval_Status = "Approved";
}
else if(input.Finance_Decision == "Reject")
{
    input.Approval_Status = "Rejected by Finance";
}
if(input.Finance_Decision == "Approve" || input.Finance_Decision == "Reject")
{
    insert into Approval_History
    [
        Purchase_Request_Lookup=input.ID
        Approval_Stage="Finance Review"
        Approver=zoho.loginuser
        Action=input.Finance_Decision
        Action_Date=zoho.currenttime
        Comments=input.Finance_Comments
        Previous_Status=oldStatus
        New_Status=input.Approval_Status
        Added_User=zoho.loginuser
    ]
}
```

## C.5 Purchase_Request – Decision Field Access (DS-05)

***// DS-05 | Form: Purchase_Request | Event: On Load***

```
// HOD_Decision: visible to the HOD user profile and the HOD portal profile
if(!thisapp.permissions.isUserInProfile("HOD") &&
!thisapp.portal.isUserInProfile("HOD"))
{
    hide HOD_Decision;
}
// Finance_Decision: visible to the Finance user profile and the Finance Portal
profile
if(!thisapp.permissions.isUserInProfile("Finance") &&
!thisapp.portal.isUserInProfile("Finance Portal"))
{
    hide Finance_Decision;
}
```

## C.6 HOD Approval Process – Actions

***// Approval process "HOD Approval" | Form: Purchase_Request | Level 1***

```
on approve:
    sendmail
    [
        from : zoho.loginuserid
        to : approver.submitter
        subject : "Approval Request"
        message : "Your request has been approved."
    ]
    update Purchase_Request[ID == input.ID]
    [
        HOD_Decision = "Approve"
        Approval_Status = "Pending Finance Approval"
    ]
on reject:
    sendmail
    [
        from : zoho.loginuserid
        to : approver.submitter
        subject : "Approval Request"
        message : "Your request has been rejected."
    ]
    update Purchase_Request[ID == input.ID]
    [
        HOD_Decision = "Reject"
        Approval_Status = "Rejected by HOD"
    ]
```

# APPENDIX D
# USER MANUAL

## D.1 Accessing the Application

Open the application link [paste the live URL here] in Chrome or Edge, or open the Zoho Creator mobile app. Sign in with the Zoho account given by the Administrator. The left menu shows the sections that are allowed for your profile.

## D.2 Instructions for the Administrator

1. Open Department and add each department with its code, name, head, status and annual budget.
2. Open Budget and add the allocated amount for each department, financial year and budget head.
3. Open Vendor and add the supplier details.
4. Use Settings → Users and Portal User Permissions to add users and assign HOD, Finance or Requester profiles.

## D.3 Instructions for the Requester

5. Open Purchase Request, select the Department and Budget Head, choose the Priority, and enter the Purpose and the Required By Date.
6. In the Items table click Add New and enter the item name, category, quantity and estimated unit price for each item. The estimated total is calculated automatically.
7. Attach supporting documents if available and click Submit.
8. Use the Purchase report to follow the status of your requests. If the status is Revision Required, correct the request and resubmit it.

## D.4 Instructions for the HOD

9. Open Purchase Request → Pending HOD Approval (or the same list on the Procurement Dashboard).
10. Open a request, check the purpose, items and budget status, enter HOD Comments, select Approve, Reject or Revision Required in HOD Decision and click Update.

## D.5 Instructions for Finance

11. Open Purchase Request → Pending Finance Approvals.
12. Open a request, check the Budget Status and Budget List, enter Finance Comments, select Approve or Reject in Finance Decision and click Update.
13. Open Approval History to see the full record of decisions.